

Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030414

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Query 1: - CREATE DATABASE

```
CREATE DATABASE IF NOT EXISTS joins_db;  
USE joins_db;
```

Output: -

#	Time	Action	Message	Duration / Fetch
✓ 1	13:55:57	CREATE DATABASE IF NOT EXISTS joins_db	1 row(s) affected	0.031 sec
✓ 2	13:56:01	USE joins_db	0 row(s) affected	0.000 sec

Query 2: -CREATE TABLES

```
CREATE TABLE class (  
    id INT,  
    name VARCHAR(30)  
);  
  
CREATE TABLE class_info (  
    id INT,  
    address VARCHAR(30)  
);
```

Output: -

✓ 3	13:58:04	CREATE TABLE class (id INT, name VARCHA...	0 row(s) affected	0.125 sec
✓ 4	13:58:07	CREATE TABLE class_info (id INT, address V...	0 row(s) affected	0.047 sec

Query 3: -INSERT VALUES

```
INSERT INTO class VALUES  
(1, 'Yishnaani'),  
(2, 'Rahul'),  
(4, 'Priya');  
  
INSERT INTO class_info VALUES  
(1, 'HYDERABAD'),  
(2, 'DELHI'),  
(3, 'CHENNAI');
```

Output: -

✓	7	14:01:03	INSERT INTO class VALUES (1,'Aswitha'), (2,'Adhr...	3 row(s) affected	Records: 3	Duplicates: 0	Wamin...	0.031 sec
✓	8	14:01:08	INSERT INTO class_info VALUES (1,'HYDERABA...	3 row(s) affected	Records: 3	Duplicates: 0	Wamin...	0.062 sec

Query 4: - CROSS JOIN

```
SELECT *  
FROM class  
CROSS JOIN class_info;
```

Output: -

✓	17	14:04:24	SELECT * FROM class CROSS JOIN class_info L...	9 row(s) returned	0.000 sec / 0.000 sec
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Query 5: - INNER JOIN

```
SELECT *  
FROM class  
INNER JOIN class_info  
ON class.id = class_info.id;
```

Output: -

✓	18	14:08:29	SELECT * FROM class INNER JOIN class_info O...	2 row(s) returned	0.000 sec / 0.000 sec
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2	Adhritha	2	DELHI
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Query 6: -INNER JOIN

```
SELECT class.name,  
       class_info.address  
FROM class  
INNER JOIN class_info  
ON class.id=class_info.id;
```

Output: -

✓	19	14:10:36	SELECT class.name, class_info.address FRO...	2 row(s) returned	0.000 sec / 0.000 sec
address					
HYDERABAD					
DELHI					

Query 7: - NATURAL JOIN

```
SELECT *
FROM class
NATURAL JOIN class_info;
```

Output: -

✓	19	14:10:36	SELECT class.name, class_info.address FRO...	2 row(s) returned	0.000 sec / 0.000 sec
✓	20	14:12:24	SELECT * FROM class NATURAL JOIN class_inf...	2 row(s) returned	0.000 sec / 0.000 sec
address					
HYDERABAD					
DELHI					

Query 8: - LEFT OUTER JOIN

```
7 • INSERT INTO class VALUES
8 (5, 'Nihitha');
9 • INSERT INTO class_info VALUES
0 (7, 'BENGALURU'),
1 (8, 'PUNE');
```

Output: -

✓	21	14:14:30	INSERT INTO class VALUES (5,'Nihitha')	1 row(s) affected	0.031 sec
✓	22	14:14:36	INSERT INTO class_info VALUES (7,'BENGALU...	2 row(s) affected Records: 2 Duplicates: 0 Wami...	0.046 sec

Query 9: - LEFT JOIN

```
SELECT *
FROM class
LEFT OUTER JOIN class_info
ON class.id=class_info.id;
```

Output: -

✓	23	14:16:23	SELECT * FROM class LEFT OUTER JOIN class...	4 row(s) returned	0.000 sec / 0.000 sec
---	----	----------	--	-------------------	-----------------------

id	address
1	HYDERABAD
2	DELHI
NULL	NULL
NULL	NULL

Query 10: - LEFT JOIN

```
SELECT *
FROM class
LEFT JOIN class_info
ON class.id=class_info.id
WHERE class_info.id IS NULL;
```

Output: -

✓ 24 14:18:46 SELECT * FROM class LEFT JOIN class_info ON... 2 row(s) returned 0.000 sec / 0.000 sec

	id	name	id	address
▶	4	Amulya	NULL	NULL
	5	Nihitha	NULL	NULL

Query 11: - RIGHT JOIN

```
SELECT *
FROM class
RIGHT OUTER JOIN class_info
ON class.id=class_info.id;
```

Output: -

✓ 25 14:20:51 SELECT * FROM class RIGHT OUTER JOIN clas... 5 row(s) returned

id	address
1	HYDERABAD
2	DELHI
3	CHENNAI
7	BENGALURU
8	PUNE

Query 12: -UNION

```
CREATE TABLE first_table(
id INT,
name VARCHAR(30)
);
CREATE TABLE second_table(
id INT,
name VARCHAR(30)
);
```

✓	28	14:25:46	CREATE TABLE first_table(id INT, name VARCH...	0 row(s) affected	0.047 sec
✓	29	14:25:50	CREATE TABLE second_table(id INT, name VA...	0 row(s) affected	0.031 sec
✓	30	14:25:54	INSERT INTO first_table VALUES (1,'Akshaya'), (...	2 row(s) affected Records: 2 Duplicates: 0 Wami...	0.046 sec
✓	31	14:25:58	INSERT INTO second_table VALUES (2,'Manasv...	2 row(s) affected Records: 2 Duplicates: 0 Wami...	0.047 sec

```
SELECT * FROM first_table
UNION
SELECT * FROM second_table;
```

✓	32	14:27:30	SELECT * FROM first_table UNION SELECT * FR...	3 row(s) returned	0.031 sec / 0.000 sec
---	----	----------	--	-------------------	-----------------------

Result Grid		
	id	name
▶	1	Akshaya
	2	Manasvi
	3	Ramya

Query 13: - UNION ALL

```
SELECT * FROM first_table
UNION ALL
SELECT * FROM second_table;

SELECT first_table.*
FROM first_table
INNER JOIN second_table
ON first_table.id = second_table.id;
```

Output: -

✓	34	14:32:12	SELECT first_table.* FROM first_table INNER JOI...	1 row(s) returned
---	----	----------	--	-------------------

Result Grid		
	id	name
▶	2	Manasvi

Query 15: -MINUS (MYSQL DO NOT SUPPORT MINUS DIRECTLY)

```
SELECT first_table.*
FROM first_table
LEFT JOIN second_table
ON first_table.id = second_table.id
WHERE second_table.id IS NULL;
```

Output: -

35 14:34:55 SELECT first_table.* FROM first_table LEFT JOIN... 1 row(s) returned

	id	name
▶	1	Akshaya

Query 16: -Generate a report showing all students and address availability status.

```
SELECT c.id,
       c.name,
       CASE
         WHEN ci.address IS NULL
         THEN 'Address Missing'
         ELSE 'Address Available'
       END AS Status
FROM class c
LEFT JOIN class_info ci
ON c.id=ci.id;
```

Output: -

37 14:37:51 SELECT c.id, c.name, CASE WH... 4 row(s) returned 0.000 s

Status
Address Available
Address Available
Address Missing
Address Missing

Query 17: -FULL OUTER JOIN (MYSQL DO NOT DIRECTLY SUPPORT FULL OUTER JOIN)

```

SELECT *
FROM class
LEFT JOIN class_info
ON class.id = class_info.id
WHERE class_info.id IS NULL
UNION
SELECT *
FROM class
RIGHT JOIN class_info
ON class.id = class_info.id
WHERE class.id IS NULL;

```

Output: -

✓	38	14:41:25	SELECT * FROM class LEFT JOIN class_info ON...	5 row(s) returned	0.000 sec / 0.000 sec
	id	name	id	address	
▶	4	Amulya	NULL	NULL	
	5	Nihitha	NULL	NULL	
	NULL	NULL	3	CHENNAI	
	NULL	NULL	7	BENGALURU	
	NULL	NULL	8	PUNE	